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Country Energy Report

June 2026

Czechia



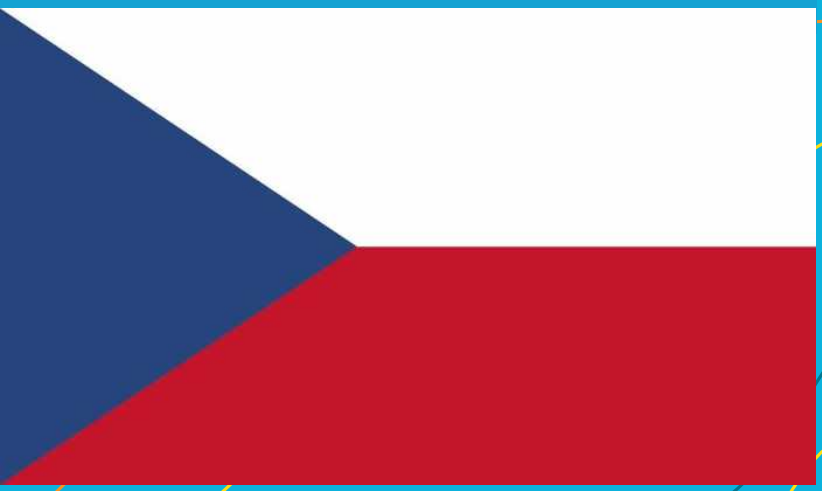


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Overview



Map sources: ESRI, Natural Earth, UNGIS. The boundaries and names shown and the designations used on this map do not imply official endorsement or acceptance by the United Nations. Map created in Jan 2025

Highlights

- Czechia aims to raise the share of renewables in final energy consumption to 30% in 2030.

  The country will phase out coal for electricity and heat generation by 2033.

- ČEZ dominates the electricity market with over 60% of total power generation and distribution.

  Net4Gas, the gas transmission company, was acquired by CEPS in 2023.

- Nuclear (42%) overpassed lignite in the power mix (35%) for the first time.

  Norway became the first gas supplier in 2023.

- Since 2021, final consumption has declined by 3%/year.

- The country plans to halve the share of coal and lignite in the energy mix in 20 years, to the advantage of nuclear (up to 4 new nuclear reactors), solar, and wind.

  The country is strengthening its gas interconnections.

- Net4Gas is involved in the "Central European Hydrogen Corridor" project with Germany, Slovakia, and Ukraine.

68%

of nuclear in the power mix in 2040

30%

share of renewables in final consumption in 2030

-50%

in GHG emissions since 1990

Table 1 : Economic Indicators

		1990	2000	2010	2019	2022	2023	2024	2025
Population	*million*	10.3	10.3	10.5	10.7	10.7	10.9	10.9	10.9
GDP growth rate	*%/year*	-2.4	4.0	2.7	3.6	2.8	-0.06	1.1	2.5
GDP/capita	*USD*	n.a.	6 062	20 160	24 063	28 282	31 591	31 707	35 639
Inflation rate	*%/year*	n.a.	3.8	1.5	2.8	15.1	10.7	2.4	2.5
Exchange rate	*lc/USD*	n.a.	38.6	19.1	22.9	23.4	22.2	23.2	21.9
Energy security		**1990**	**2000**	**2010**	**2019**	**2022**	**2023**	**2024**	**2025**
Energy independence rate	*%*	83	75	71	63	61	61	58	59
Share of oil imported(+) exported(-)	*%*	97	95	97	98	99	99	99	99
Energy efficiency		**1990**	**2000**	**2010**	**2019**	**2022**	**2023**	**2024**	**2025**
Total consumption/GDP *	*koe/USD15p*	0.213	0.169	0.136	0.104	0.100	0.091	0.088	0.087
Total consumption/GDP *	*2015=100*	182	144	116	88.4	85.1	77.3	75.3	74.7
Rate of T&D power losses	*%*	7.0	8.7	7.1	6.6	5.4	5.0	5.2	5.1
Efficiency of thermal power plants	*%*	27.8	31.8	30.0	33.7	33.9	33.4	33.5	33.4
CO_2 emissions		**1990**	**2000**	**2010**	**2019**	**2022**	**2023**	**2024**	**2025**
CO_2 emissions/GDP *	*kCO_2/USD15p*	0.651	0.489	0.334	0.228	0.214	0.189	0.174	0.173
CO_2 emissions/capita	*tCO_2/cap.*	14.7	11.7	10.6	8.8	8.4	7.3	6.8	6.9

** at purchasing power parity*

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Energy & Climate Policy Framework



Czechia aims to raise the share of renewables in final energy consumption to 30% in 2030.

The country will phase out coal for electricity and heat generation by 2033.

The **Government Council for the Energy and Raw Materials Strategy** of the **Ministry of Industry and Trade** is in charge of the country's energy policy.

The **Energy Regulation Office (ERO)** has overseen energy sector regulation since 2001. In 2011, it took over many competences of SEI, the State Energy Inspection, including the control of energy and heat prices and compliance with the Energy Act.

The National Energy and Climate Plan (NECP, 2024) set energy and GHG targets for the period 2021–2030 with a view to 2050.

A CZK300bn (€12.3bn) modernisation fund, financed by the sale of carbon allowances, was created in 2021 to invest in renewables (39%), modernize heat supply systems (26%), improve energy efficiency, and cut GHG emissions from companies subject to the EU ETS (13%).

In 2015, the government approved its energy strategy (ASEK), which includes six scenarios for development of the energy sector by 2040. Nuclear power should be a pillar of the energy strategy, with its share in the power mix rising from about 40% in 2023 to between 46% and 58% by 2040. The targets have since been updated in the country's 2024 NECP, projecting a nuclear share of 44% by 2030 and 68% by 2040.

The Energy Act (2000) was amended in 2016 to fully implement the EU's Third Energy Package and enhance consumer protection. A new amendment was approved in 2025 (Lex OZE III) to accelerate renewable energy deployment, recognise energy storage, and enhance market flexibility.

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Energy Market Regulations

ELECTRICITY

The electricity market has been open to all consumers since 2006 and to all professionals since 2005. The opening process started in 2002 for consumers over 40 GWh/year (30% of the market), and in 2003 for consumers over 9 GWh/year (40% of the market).

In 2016, amendments to the Energy Act removed high contract termination penalties, which led to a 60% surge in customers' switches between 2016 and 2018. The number of supplier switches increased by 3% in 2025 to 596 000; it reached a peak of over 1 million in 2021, following the failure of the Bohemia Energy Group that left 900 000 electricity and gas consumers without a supplier.

In 2019, Czechia joined the Single Intraday Coupling (SIDC), which aims to create an integrated European intraday market and ensure the continuous trading of power across 25 countries. In 2021, the country, along with Slovakia, Hungary, and Romania, coupled its day-ahead markets with the Pan-European day-ahead power market.

OTE (100% state-owned), the electricity market operator, organises the short-term electricity market and the system adequacy in cooperation with the operator of the transmission system. Most exchanges on the electricity market take place under bilateral contracts (more than 90% of power purchases). In 2025, 32 TWh (-1%) were traded on OTE's day-ahead market at an average price of €97/MWh (-16%).

PXE (Power Exchange Central Europe), part of the EEX Group (European Energy Exchange, 66.67%), is involved in the exchange of "futures" in Czechia, Slovakia, Hungary, Romania, and Poland. In 2025, 270 TWh of power futures (with financial settlement) were traded (+36%), including 26 TWh on the Czech market.

In April 2026, the government announced the results of the second 2025 auction aimed at awarding 910 MW of high-efficiency combined heat and power (CHP) generation capacity. Nearly 580 MW were awarded across 67 projects, with commissioning scheduled by 2030. The supported fuel types include natural gas, biomass, biogas, and coal gas. For natural gas projects, applicants must demonstrate (in line with EU state aid rules) that the facilities are

technically and economically capable of replacing natural gas and operating on renewable or low-carbon gases by 2050.

In 2024, the European Commission approved a CZK75bn (€3.2bn) scheme to support the production of electricity from new and modernised high-efficiency CHP plants, except for those powered by solid fossil fuels, diesel, and oil. Under the scheme, the aid will take the form of a feed-in premium per MWh for a duration of 15 years. The bonus is set through tenders, except for small installations (up to 1 MWe) where the amount is set administratively by ERU, the Regulatory Office, on an annual basis and limited to the funding gap. The scheme remained in force until the end of 2025.

KEY OBSERVATIONS

The electricity market is open to all customers since 2006.

GAS

The gas market has been open to all consumers since mid-2007. The opening started in 2005 for customers consuming more than 15 mcm/year (28% of the market) and to all professionals in 2006 (140 000 consumers). As for electricity, switch penalties were removed in 2016, reversing the declining trend in supplier switching since 2013. The number of switches reached205 000 in 2025 (stable compared to 2024).

PXE is also the gas stock exchange. In 2025, over 36 TWh of gas were traded (with physical settlement) (+3% compared to 2024), including 25 TWh on the spot market (+26%) and 11 TWh on the future market (-26%).

COAL

The coal and lignite sector is facing some difficulties due to relatively high production costs, mainly for hard coal. In addition, the sector is facing stricter EU climate policies and rising EU carbon prices.

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In 2022, the Government chose to advance the phase-out of coal to 2033 (from 2038), while relying more on nuclear energy. This objective was once again set in the update of the NECP.

Low Carbon Options

RENEWABLES

The final updated NECP (2024) aims to raise the share of renewables in final energy consumption to 30% in 2030. The share of RES in electricity generation is projected to increase to 28% by 2030 and to 46% by 2050. In the heating sector, the share of RES is expected to be 40% in 2030, eventually reaching 74% by 2050. In the transport sector, Czechia has opted for the GHG intensity reduction pathway under RED III, committing to a 14.5% reduction in the carbon intensity of transport fuels by 2030.

The target set for 2020 of 13.5% of renewables in final energy consumption has been exceeded by almost 4 points (17.3%).

To achieve the 30% share of RES in final energy consumption, plans include the installation of an additional 8 GW of solar PV capacity and 1.2 GW of wind power capacity connected to the grid by 2030 (NECP).

KEY OBSERVATIONS

The share of RES in power generation is expected to reach 28% by 2030.

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EnerFuture: Country-level Energy Forecasts



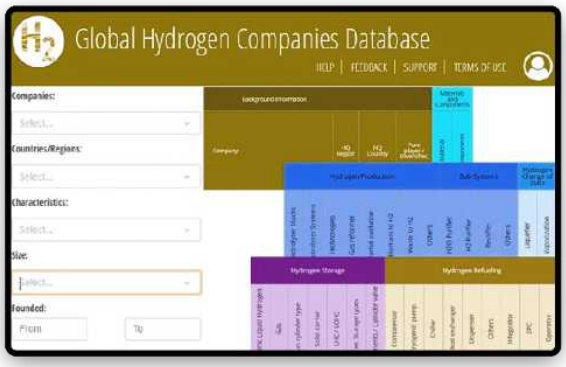
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